

# Air Quality Index (AQI) Analysis of Delhi

## Introduction:

Air pollution has emerged as one of the biggest environmental problems in urban cities like Delhi. The Air Quality Index (AQI) is a number-based method employed to gauge and communicate air quality and their impacts on humans.

AQI takes various components into account, including PM<sub>2.5</sub>, PM<sub>10</sub>, NO<sub>2</sub>, CO, SO<sub>2</sub>, and O<sub>3</sub>, among others, and assigns them a score for measuring air quality. Higher AQI scores reflect higher levels of air pollution and related adverse impacts on humans.

Air pollution is common in Delhi owing to various reasons, including automobile emissions, industrial operations, burning of crops, and weather conditions. The present project intends to conduct an AQI analysis and propose improvements in this regard.

## Objectives and Research Questions:

The main objectives of the research include the following:

- To examine the temporal trend of AQI in Delhi.
- To detect the pollutants influencing AQI.
- To observe the seasonal trends in AQI levels.
- To investigate the monthly and annual trends in pollution.
- To examine the correlation among various pollutants.

## Research Questions:

1. Which pollutant is responsible for the maximum contribution in AQI in Delhi?
2. Does AQI change according to the season?
3. In which month do pollution levels peak?
4. Are there any correlations between different pollutants?
5. Which environmental factors affect AQI in Delhi?

## Dataset Description:

The dataset used for this analysis contains air quality measurements collected from monitoring stations in Delhi.

### **Dataset Features:**

- **Date** → Date of observation
- **PM2.5** → Fine particulate matter
- **PM10** → Coarse particulate matter
- **NO2** → Nitrogen dioxide
- **CO** → Carbon monoxide
- **SO2** → Sulphur dioxide
- **O3** → Ozone
- **AQI** → Air Quality Index

The dataset provides time-based pollution data, which helps in analysing trends, seasonal variations, and pollutant impact.

### **Methodology:**

The analysis was carried out using Python with libraries such as Pandas, Matplotlib, and Seaborn.

#### **Steps followed:**

##### **1. Data Collection:**

The data set was collected from the given source.

##### **2. Data Cleaning:**

Missing data was deleted or imputed.

Column “Date” was changed to its correct format.

##### **3. Feature Engineering:**

Month and year were extracted from “Date”.

A new column for Season was created.

##### **4. Data Analysis:**

Mean and trends were computed.

Months and seasons wise data was grouped.

##### **5. Visualizations:**

Trends of AQI by line plot.

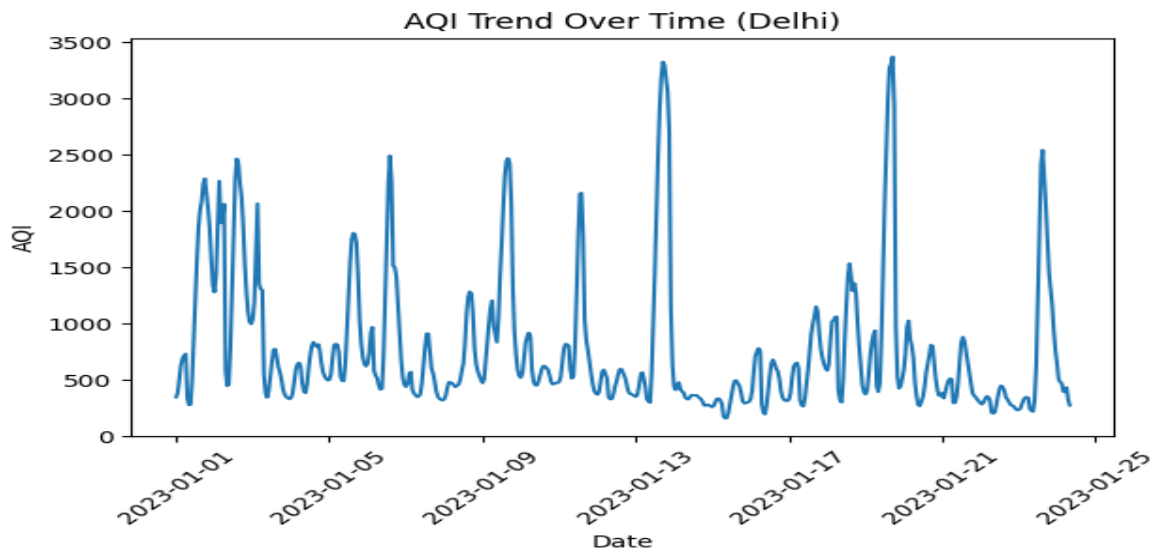
Analysis by months with bar graph.

Seasons analysis using boxplot method.

Heat map for correlations analysis.

## Visualizations and Explanation:

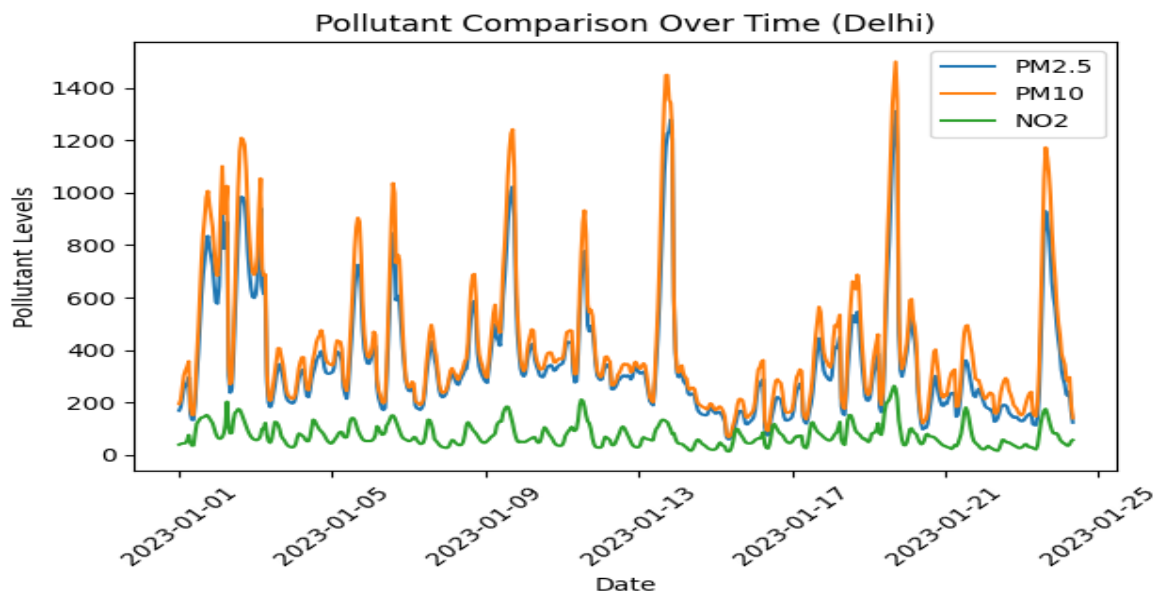
### 1. AQI Trend Over Time



#### Explanation:

The line graph shows how the AQI changes over time. It helps find times when pollution is high and general trends. A trend that is going up means that the air quality is getting worse.

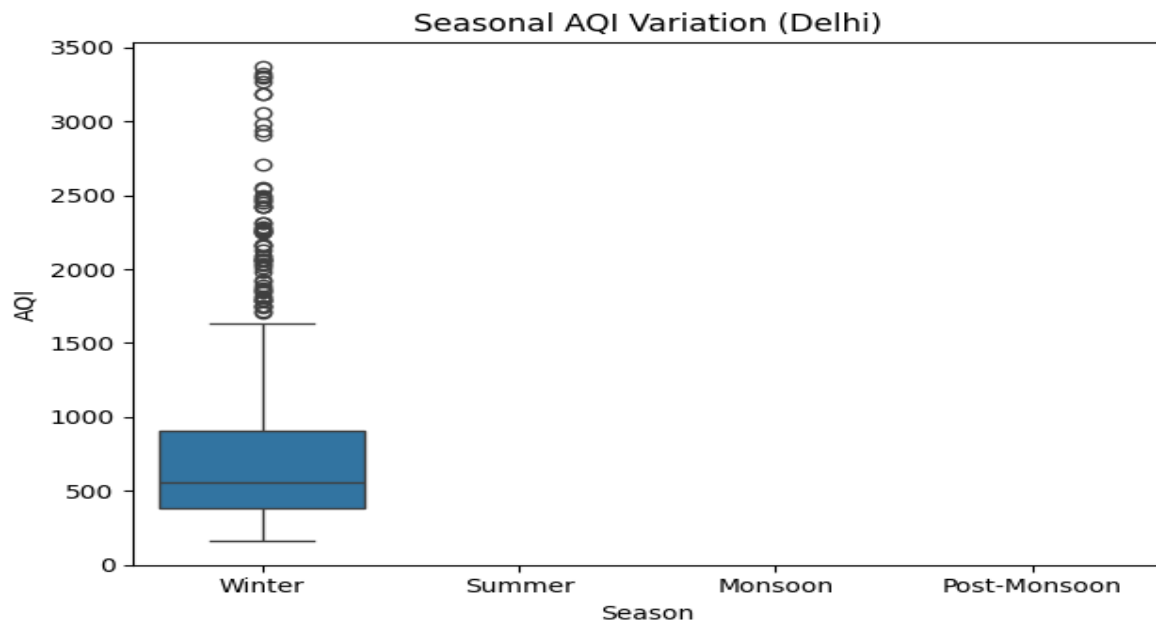
### 2. Pollutant Comparison



#### Explanation:

This graph compares different pollutants such as PM2.5, PM10, and NO2. It helps identify which pollutant contributes most to air pollution.

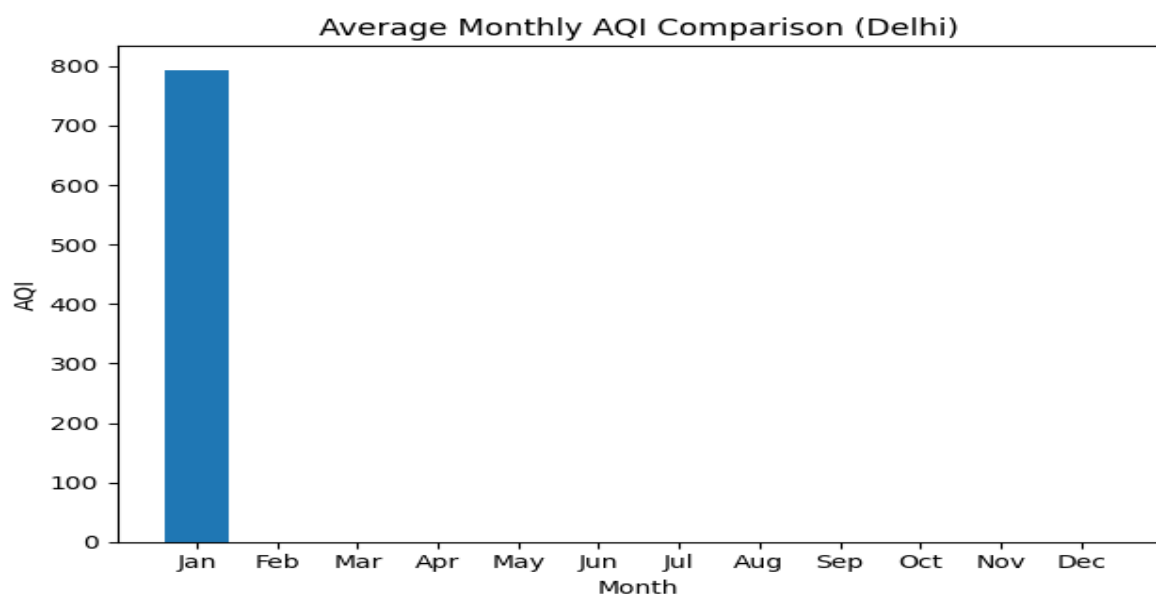
### 3. Seasonal AQI Analysis



#### Explanation:

The boxplot shows AQI variation across seasons. Typically, AQI is highest in winter due to fog and pollution trapping, while it is lower during monsoon and other seasons.

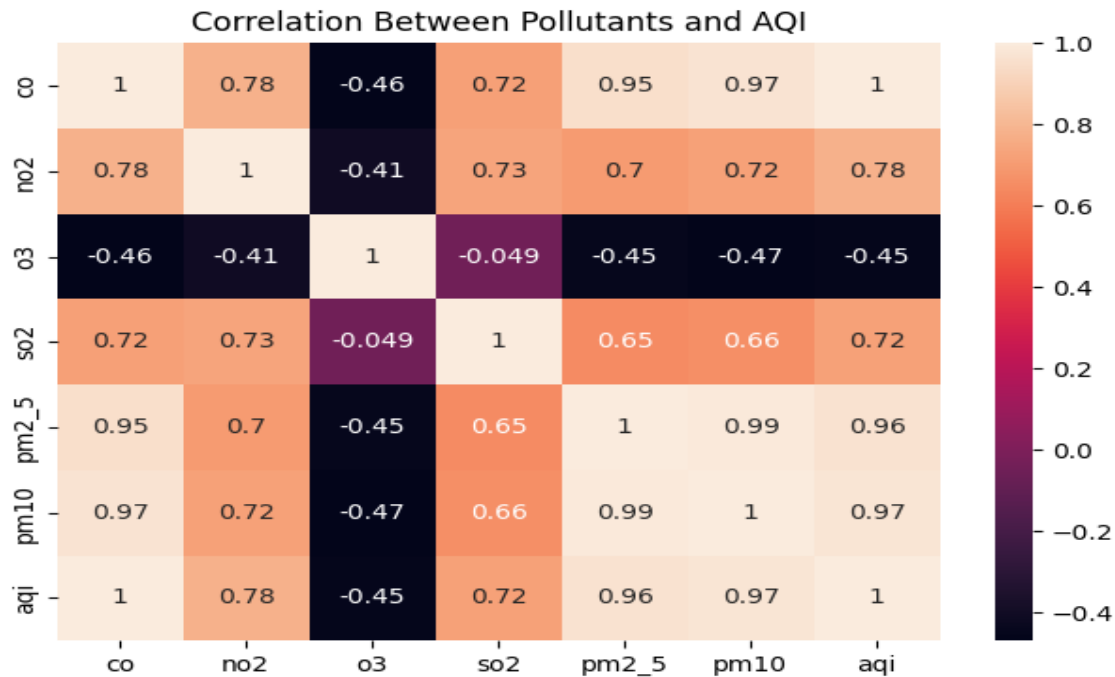
### 4. Monthly AQI Analysis



#### Explanation:

The bar chart is an illustration of the mean AQI values per month taken from the dataset. This makes it possible to compare air quality throughout the months and determine the months that experience more pollution in Delhi.

## 5. Correlation Heatmap



### Explanation:

The heatmap shows relationships between pollutants. Strong correlation between PM2\_5 and AQI indicates its major impact on air quality.

### Analysis & Insights

Based on the results of the analysis, it can be concluded that:

- The AQI is relatively high during winters.
- PM2.5 and PM10 cause most pollution.
- There is an increase in pollution from October to January.
- There is a high relationship between PM2.5 and AQI.
- There is a better air quality during monsoons due to rain.

### Environmental Challenges in Delhi

Environmental issues that impact air pollution in Delhi include:

- Emissions from vehicles due to increased vehicular traffic.
- Pollution caused by industries.
- Burning of crops in surrounding states.
- Dust from construction sites and roads.
- Weather-related factors such as low windspeed and temperature inversion.

## **Suggestions & Solutions**

To enhance air quality, one can undertake the following:

- Encourage the use of public transport and electric vehicles.
- Enforce industrial emissions standards.
- Regulate agricultural burning practices.
- Plant trees and increase vegetation cover.
- Enhance pollution prevention knowledge among citizens.

## **Conclusion**

AQI data of Delhi shows that the problem of air pollution is quite significant, particularly in winter. The pollutants like PM<sub>2.5</sub> contribute significantly to poor air quality in Delhi.

Such knowledge can be helpful for creating strategies to minimize pollution. It becomes vital to develop appropriate measures for improving air quality in Delhi.